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Form no. - 1002107305



Chemistry Test

Question 1

1. (4) Argon
2. (3) Lithium
3. (3) Fluorspar
4. (3) 16
5. (1) $\text{Ag} + \text{e}^- \rightarrow \text{Ag}^+$
6. (2) $\text{Cu}(\text{OH})_2$
7. (4) Benzene
8. (2) conc. sulphuric acid
9. (1) Lead Nitrate
10. (3) KHSO_4
11. (4) Non-volatile acid property

12 (2) Ethane

13 (3) H_2SO_4

14 (2) C_nH_{2n}

15 (4) Phenolphthalein

16

(a) Sodium chloride

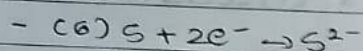
- (2) Has high melting
and boiling point

(b) Methane - (3) Greenhouse gas

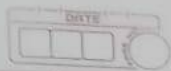
(c) Hydrogen chloride gas

- (4) Has low melting
and boiling point

(d) Oxidation Reaction



(e) Water - (1) Has two shared pair
of electrons.



(ii)

(a) The constituent of electrolyte mixture is - Fluorspar, cryolite, Al_2O_3 .

(b) coke powder

(c) Hall Herault's Process

(d) Anode - $3\text{C} + 6\text{e}^- \rightarrow 3\text{CO} \rightarrow 3\text{CO}_2$
 $\text{C} + 2\text{O}^{2-} \rightarrow \text{CO}_2 + 2\text{e}^-$

cathode - $2\text{Al}^{3+} + 6\text{e}^- \rightarrow 2\text{Al}$

(iii)

(a) reducing agents

(b) bad

(c) Lead chloride alkaline

(d) Lead chloride

(e) Hydrogenation

(iv)

(a) Electron Affinity

(b) Catenation

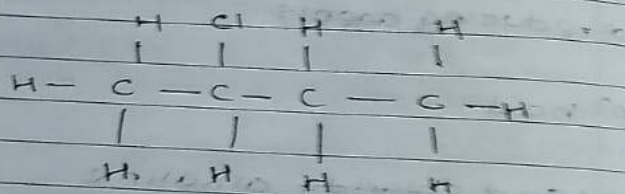
(c) Haber's Process

(d) Basic Salt

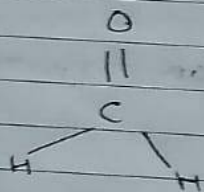
(e) Vapour Density

(v)

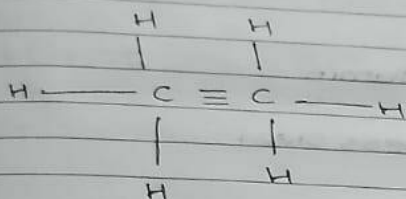
(a) 2-chlorobutane



(2) (b) Methanal



3



(b)

1. Ethanoic acid

2. Butanol

Q8

(i)

(a) Hydrogen Chloride Gas is highly reactive and can easily react with quicklime (CaO). Hence, to prevent the reaction, the gas is not dried over CaO.

(b) Ammonia is not collected over water because it is highly soluble in water.

(i)

(a) duralumin

(b) Stainless steel



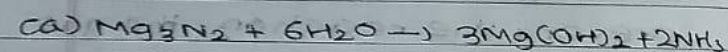
2 vol 7 vol 4 vol 6 vol

80

2(40) 7(40) 4(40) 6(40)

Resultant mixture's volume
= 160 ml

(iii)



(b) since, nitrides are expensive

this method is seldom used.

cc) On bringing HCl near the gas, dense white fumes are formed, which proves the gas formed is NH_3 .

Q19

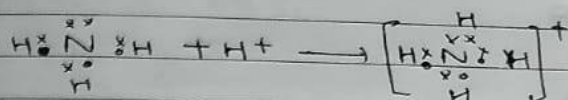
(i)

(a) Bronze is used in making statues, medals, coins.

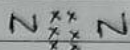
(b) Fuse metal is used for electrical fuse and soldering purposes.

(ii)

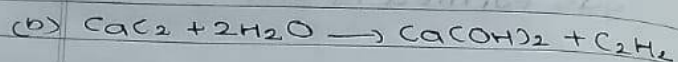
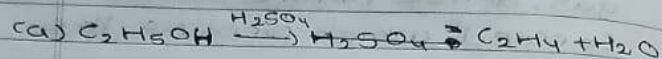
(a) Ammonium Ion



(b) $\text{N} \equiv \text{N}$



(iii)



(iv)

(a) Nitrate

(b) Sulphide

(c) Sulphate

Q20

(1)

(a) Electronegativity - tendency of an atom to attract electrons towards itself.

(b) Gay Lussac's Law - When gases react, they do so in volumes

and bear a simple whole number ratio to one another and to the product of their masses.

(ii) Carbon = 14.4%
weight = 12

Relative atoms = $\frac{14.4}{12} = 1.2$

ratio = 1

Hydrogen = 1.2%
weight = 1

Relative atoms = 1.2
ratio = 1

Chlorine = 84.5%
weight = 35.5

relative atoms = 2.38

ratio = 2

empirical formula = CHCl_2

Molecular mass

$= 168$

$= 83$

$= 2$

Molecular formula $= C_2H_2Cl_4$

(iii)

(a) Copper

(b) Sodium sulphate

(c) Ferrous sulphate

(iv)

(a) Pale blue solution becomes inky blue.

(b) NH_3 burns in atmosphere of excess oxygen with greenish yellow flame, forming nitrogen and water vapour.

(c) Buff yellow PbO is reduced to greyish metallic red.

(d)

(i)

(a) Isomers

(b) Homologous series

(ii)

(a) Ionization potential decreases with increase in atomic size. Down the group Atomic size increases therefore Ionization potential decreases.

(b) Ionic compounds can't conduct electricity in solid state due to absence of free ions.

(iii) Mol. mass = $40 + 2(2 + 31 + 64)$
= 234 g

Mass of fertilizer = $2(31)$
= 62

% of P = $\frac{62}{234} \times 100 = 26.5\%$

(b) C_4H_9

(iv)

(a) Impure copper

(b) Pure copper is deposited on the thin sheet of pure copper placed at cathode.

